How Timer makes a forecast

Tsinghua University (THUML) · 84M parameters · GPT-style decoder, next-patch prediction · zero-shot

1. One series at a time

Timer reads only the target's own past. A ? is a value not known yet.

past *future*

Target temperature

It needs at least 96 steps; older steps are dropped so the length is a multiple of 96.

No covariates; each metal on its own.

2. Scale, then patches of 96

The context is scaled by its mean μ and standard deviation σ, then cut into patches of 96 steps.

$$\tilde{x} = \frac{x - \mu}{\sigma}$$

context = 6 patches of 96

linear layer

6 tokens of 1,024 numbers

3. Eight causal blocks

Each patch reads only itself and earlier patches, as in a GPT language model. LayerNorm in front, shortcut around.

earlier → later

earlier patches only

Self-attention: causal, 8 heads, rotary positions (RoPE)

Feed forward: SwiGLU, 1,024 → 2,048 → 1,024

4. The next 96 steps

A linear head turns the last token into the **next 96 values**. A longer horizon appends them and runs again.

point forecast

Timer gives a point forecast only, with no intervals.

Forecasting as next-token prediction: the GPT recipe applied to patches of a series.

8 blocks · width 1,024 · 8 heads · patches of 96 in and out

In the notebooks: last 2,048 days or full monthly history · each metal on its own · no covariates · every horizon in one pass